

UX-0011 — Butlerly Design Patterns and Component Library

Version 1.0

Purpose

This document defines Butlerly's reusable interaction patterns and production component library for Version 1 across smartphone, tablet, and desktop. It converts the design foundations, approved flows, screen inventory, AI behavior, cross-device continuity, accessibility standard, and wireframe requirements into consistent, testable building blocks for design and engineering.

1. Objectives

The library must:

- Make common behaviors consistent across all three native experiences.
- Prefer platform-native controls when they satisfy Butlerly requirements.
- Separate semantic intent from platform presentation.
- Make privacy, AI, synchronization, recovery, and accessibility first-class component behavior.
- Reduce duplicated design and implementation work.
- Preserve one product language while respecting device-specific conventions.
- Include complete state, accessibility, analytics, and test contracts.

2. Governing Principles

2.1 Meaning Before Appearance

Components are defined by user purpose, behavior, state, and consequence before visual styling.

2.2 Native First, Custom by Exception

Use native controls for navigation, forms, dialogs, menus, focus, text scaling, authentication, sharing, and accessibility unless a documented requirement cannot be met. Custom controls require design and engineering review.

2.3 One Semantic Component, Multiple Native Renderings

A semantic component may render differently on iPhone, iPad, and desktop. Its meaning, data contract, permissions, accessibility outcome, and state model remain consistent.

2.4 Complete States Are Part of the Component

Default, hover, focus, pressed, selected, disabled, loading, empty, offline, degraded, error, success, and recovery behavior are defined where applicable.

2.5 Trust-Critical Behavior Is Never Decorative

Privacy, security, AI confidence, device trust, data location, synchronization, destructive consequences, and recovery cannot depend on color, animation, or icons alone.

2.6 Composition Over Duplication

Screens are assembled from reusable primitives, components, and patterns. New components are created only when existing ones cannot express the required behavior without distortion.

3. Library Architecture

The library has five layers:

Layer A — Foundations

Semantic color, typography, spacing, shape, elevation, iconography, motion, breakpoints, and accessibility tokens defined by UX-0007.

Layer B — Primitives

Text, icon, divider, spacer, surface, image, focus ring, progress indicator, and status mark.

Layer C — Controls

Button, text field, selection control, menu, tab, disclosure, search field, date input, amount input, and file picker.

Layer D — Composites

Record row, review card, AI explanation, sync status, device row, source preview, filter bar, table, form section, empty state, and error state.

Layer E — Product Patterns

Capture, review, consent, action preview, device pairing, conflict resolution, export, deletion, recovery, and cross-device continuation.

4. Component Naming and IDs

Format:

C-CATEGORY-NNN

Categories:

- PR — primitive.
- CT — control.
- NV — navigation.
- DT — data display.
- FB — feedback.
- AI — AI interaction.
- PV — privacy and security.
- SY — synchronization and device continuity.
- RC — record and capture.
- PT — composed product pattern.

Examples:

- BL-CT-PRIMARY-BUTTON
- BL-DT-RECORD-ROW
- BL-AI-EXPLANATION-CARD
- BL-SY-DEVICE-ROW

- BL-PT-DESTRUCTIVE-CONFIRMATION

Variants are documented under the canonical C-CATEGORY-NNN component ID. Descriptive implementation aliases may be used in code, but BDS component IDs remain authoritative.

5. Required Component Contract

Every production component must document:

- Purpose and when to use it.
- When not to use it.
- Anatomy.
- Content rules.
- Variants and sizes.
- Inputs, outputs, and events.
- State model.
- Platform adaptations.
- Keyboard and pointer behavior.
- Screen-reader name, role, value, state, and announcements.
- Touch target and text scaling behavior.
- Light, dark, and high-contrast behavior.
- Reduced-motion behavior.
- Localization and right-to-left behavior.
- Privacy and security considerations.
- Analytics events excluding private content.
- Automated and manual test coverage.
- Dependencies and version status.

6. Core Controls

6.1 Buttons — BL-CT-BUTTON

Variants:

- Primary — one main action per decision area.
- Secondary — meaningful alternate action.
- Tertiary — low-emphasis action.
- Destructive — irreversible or high-consequence action.
- Icon — compact action with accessible name.
- Menu — opens a set of actions.

Rules:

- Labels use verbs and name the outcome.
- Disabled buttons must explain unmet requirements nearby.
- Loading preserves the button width and announces progress.
- Destructive actions use explicit labels such as “Delete permanently,” not “Continue.”
- Icon-only use is limited to well-established platform symbols and always includes an accessible name.

States:

Default, hover where applicable, focused, pressed, disabled, loading, success acknowledgment, and error recovery.

6.2 Text Field — BL-CT-TEXT-FIELD

Includes persistent label, input, optional hint, validation message, and supporting action.

Variants:

- Single line.
- Multiline.
- Search.
- Secure input.
- Read-only.
- AI-proposed value.

Rules:

- Placeholder is never the sole label.
- User input is preserved after validation or provider failure.
- AI-proposed values are visibly and programmatically identified, editable, and linked to evidence where relevant.

6.3 Amount Input — BL-CT-AMOUNT-INPUT

Displays locale-aware currency, decimal behavior, sign, and formatting without changing the underlying value unexpectedly. Gains and losses never rely only on color.

6.4 Date and Time Input — BL-CT-DATE-TIME

Uses platform-native pickers where practical. Manual entry remains possible where accessibility or workflow requires it. Locale, timezone, and source timestamp are explicit when consequential.

6.5 Selection Controls — BL-CT-SELECTION

Includes checkbox, radio, switch, segmented control, and multi-select row.

Rules:

- Switches apply immediate reversible settings.
- Radio groups select one option before submission.
- Checkboxes select independent items.
- No consent is preselected.
- Selected state is exposed visually and programmatically.

6.6 Search Field — BL-CT-SEARCH

Supports clear, cancel, recent searches, suggestions, keyboard shortcut, and local/offline scope. It must not silently convert a normal search into AI processing.

7. Navigation Components

7.1 Smartphone Bottom Navigation — BL-NV-BOTTOM-NAV

Destinations: Home, Transactions, Review, Search, Settings. Capture remains a prominent global action without obscuring navigation. Assistant is optional and non-primary in Finance V1.

7.2 Adaptive Sidebar — BL-NV-SIDEBAR

Used on tablet and desktop. Supports selected state, badges only for actionable counts, collapse behavior, keyboard navigation, and large-text adaptation.

7.3 Top App Bar / Toolbar — BL-NV-TOOLBAR

Contains screen title, contextual actions, Search, Capture or Import, and status access. Actions move to overflow based on priority and width.

7.4 Breadcrumb / Context Path — BL-NV-CONTEXT-PATH

Desktop-only where hierarchy requires orientation. It is not a replacement for a clear page title or back behavior.

7.5 Tab and Segmented Navigation — BL-NV-TABS

Used for peer views within one destination, not for unrelated workflows. Selection persists where meaningful and is exposed to assistive technologies.

8. Data Display Components

8.1 Record Row — BL-DT-RECORD-ROW

Anatomy:

- Primary identity.
- Key amount or value.
- Date or source.
- Status.
- Optional confidence or review marker.
- Optional trailing action.

Rules:

- Entire row has one clear primary target.
- Secondary actions are separately focusable and specifically named.
- Sensitive data can be masked in privacy-sensitive contexts.

8.2 Record Card — BL-DT-RECORD-CARD

Used where additional context or action is needed. It must not replace a compact row merely for decoration.

8.3 Data Table — BL-DT-TABLE

Desktop and expanded tablet pattern for high-density review.

Requirements:

- Associated headers.
- Sort and filter state.
- Keyboard row navigation.
- Explicit selection count.
- Resizable or hideable columns where supported.
- Alternate list or detail view for large text and screen readers.
- Batch actions only on explicit selection.

8.4 Key-Value Group — BL-DT-KEY-VALUE

Displays record fields in a stable reading order. Long values wrap; amounts, dates, and IDs remain understandable.

8.5 Status Badge — BL-DT-STATUS

Semantic statuses include Draft, Needs Review, Confirmed, Offline, Pending Sync, Delayed, Revoked, Recovery Needed, and Deleted. Every badge includes text; icons and color are supplemental.

8.6 Confidence Indicator — BL-AI-CONFIDENCE

Values: High, Medium, Low, or Not Available. It always includes text and, where material, a concise explanation of what lowers confidence.

9. Feedback Components

9.1 Inline Message — BL-FB-INLINE

For field or local-area feedback. Includes message, severity, and correction path.

9.2 Banner — BL-FB-BANNER

For destination-level conditions such as offline work, delayed sync, or provider degradation. It must state what is preserved and what the user can do.

9.3 Toast / Snackbar — BL-FB-TOAST

Used only for brief confirmation with no unique durable information. Consequential Undo must also have history or recovery.

9.4 Progress — BL-FB-PROGRESS

Variants: determinate, indeterminate, multi-stage, and background. Progress must not imply completion before validation. Users can safely leave long-running tasks where supported.

9.5 Empty State — BL-FB-EMPTY

Explains the purpose of the area and one or two useful next actions. It avoids shame, artificial urgency, and engagement pressure.

9.6 Error State — BL-FB-ERROR

Every error states:

- What failed.

- What was preserved.
- Whether private data left the device when relevant.
- What remains usable.
- What the user can do next.
- Whether retry is safe.

10. Overlay and Container Components

10.1 Sheet — BL-CT-SHEET

Used for focused secondary work on smartphone and tablet. Supports keyboard avoidance, drag-dismiss safeguards, accessible title, focus containment, and state preservation.

10.2 Dialog — BL-CT-DIALOG

Used for bounded decisions. It cannot contain a complex workflow that deserves a screen. Destructive dialogs place initial focus on the safest reasonable action.

10.3 Popover and Menu — BL-CT-POPOVER

Used for contextual actions and compact choices. Hover is never required. Menus close predictably and restore focus.

10.4 Pane — BL-CT-PANE

Tablet and desktop region for list-detail, source-detail, or inspector layouts. Pane titles, resizing limits, collapse behavior, and reading order are documented.

11. Record and Source Components

11.1 Source Preview — BL-RC-SOURCE-PREVIEW

Supports image, PDF, imported row, and other evidence types.

Requirements:

- Original source remains distinguishable from interpreted data.
- Zoom, rotate, page navigation, and open-original actions are accessible.
- Extracted text is available when present.
- Visual bounding boxes have text-based field relationships.
- Sensitive previews are protected in task switchers where platform support permits.

11.2 Editable Record Form — BL-RC-RECORD-FORM

Organizes fields into meaningful groups. It preserves edits, exposes AI provenance, supports source comparison, and uses inline validation plus an error summary for long forms.

11.3 Version History — BL-RC-VERSION-HISTORY

Shows timestamp, device, origin type, changed fields, and prior values. Same-field timestamp resolution and automatic merges are understandable without exposing unnecessary technical detail.

11.4 Duplicate Comparison — BL-RC-DUPLICATE-COMPARISON

Shows matched fields, differences, source evidence, recommendation, and actions: Keep Both, Merge, or Not a Duplicate. Merge previews the resulting record.

12. AI Components

12.1 AI Explanation Card — BL-AI-EXPLANATION

Structure:

1. What Butlerly believes.
2. Evidence used.
3. Confidence or uncertainty.
4. What Butlerly did or recommends.
5. What the user can do next.

Observation, inference, and recommendation are explicitly distinguishable.

12.2 AI Proposal Field — BL-AI-PROPOSAL

Displays proposed value, confidence, evidence link, Accept, Edit, and Reject where appropriate. Acceptance is never implied by passive viewing.

12.3 AI Action Preview — BL-AI-ACTION-PREVIEW

Required before consequential AI-assisted actions. Shows target, scope, changes, data use, reversibility, and confirmation.

12.4 External Processing Consent — BL-AI-CONSENT

Shows what data will leave the device, provider or processing class, purpose, expected result, retention statement when known, and alternatives. Consent is explicit and revocable.

12.5 Assistant Answer — BL-AI-ANSWER

Contains direct answer, evidence summary, uncertainty, related records, and next action. Streaming does not repeatedly disrupt assistive technology; completion is announced once.

12.6 AI Unavailable — BL-AI-UNAVAILABLE

Explains which capability is unavailable while preserving access to local records and manual workflows.

13. Privacy and Security Components

13.1 Privacy Disclosure — BL-PV-DISCLOSURE

Communicates data location, processing boundary, trusted-device scope, and user control using concise primary text with expandable detail.

13.2 Authentication Gate — BL-PV-AUTH-GATE

Explains the protected action before invoking biometric or passcode authentication. Accessible platform fallback is mandatory.

13.3 Destructive Confirmation — BL-PT-DESTRUCTIVE-CONFIRMATION

Shows object, scope, consequences, recoverability, retention period, and authentication requirement. Destructive labels are explicit.

13.4 Export Scope Selector — BL-PV-EXPORT-SCOPE

Supports all data, date range, selected records, sources, and metadata where applicable. Preview explains file contents and security implications.

13.5 Security Event Row — BL-PV-SECURITY-EVENT

Shows date, device, action, outcome, and response. Sensitive secrets never appear.

14. Synchronization and Device Components

14.1 Sync Status — BL-SY-SYNC-STATUS

States: Synced, Pending, Initial Sync, Delayed, Offline, Provider Degraded, Conflict, Recovery Needed. Healthy status is quiet; attention states are actionable.

14.2 Device Row — BL-SY-DEVICE-ROW

Shows user-defined name, platform, trust state, last successful sync, and pending-change status. Revoked devices remain identifiable in history.

14.3 Device Detail — BL-SY-DEVICE-DETAIL

Includes trust date, last activity, sync state, credential status, Rename, Troubleshoot, Lost Device, and Revoke.

14.4 Pairing Code / Verification — BL-SY-PAIRING

Supports QR plus human-readable verification phrase. Codes are short-lived and one-time. Manual alternatives are provided.

14.5 Initial Sync Progress — BL-SY-INITIAL-SYNC

Shows phases, progress, safe interruption, local usability, and integrity validation.

14.6 Conflict Comparison — BL-SY-CONFLICT

Shows competing values, devices, timestamps, evidence, recommendation, Choose, Edit, Merge, or Defer. Unrelated records remain usable.

14.7 Continue Recent Work — BL-SY-CONTINUE

Shows task name, origin device, last activity, Resume, and Dismiss. It restores semantic task context, not exact pixels.

15. Product Interaction Patterns

15.1 Capture Pattern

Launch → permission explanation when needed → acquire source → quality review → preserve source → process locally or request consent → review uncertain fields → save → Undo/history.

15.2 Review Pattern

State the issue → explain why it matters → show evidence → present one primary action → allow edit, defer, or dismiss where safe → confirm result and recovery path.

15.3 Consent Pattern

Explain purpose before asking → show exact scope → provide meaningful alternative → require explicit choice → record consent basis → permit later revocation.

15.4 Consequential Action Pattern

Initiate → preview target and changes → show privacy and reversibility → authenticate if required → execute → validate → confirm → provide Undo or durable recovery.

15.5 Offline Pattern

State offline condition quietly → preserve local work → mark pending changes → keep available functions active → identify unavailable external functions → synchronize later without duplicate work.

15.6 Degraded Provider Pattern

Identify affected provider capability → preserve local functions → state whether data left the device → offer Retry and alternative path → retain diagnostic detail separately.

15.7 Device Pairing Pattern

Choose method → authenticate → verify device identity → approve → establish keys → transfer selected encrypted data → validate → mark Active.

15.8 Merge Pattern

Different records synchronize independently. Different fields of the same record merge. Same-field changes use the latest valid timestamp in Version 1, with prior values retained. Only irreconcilable meaning or integrity issues create a review item.

15.9 Recovery Pattern

Explain what is recoverable → authenticate → validate identity and source → restore without duplicate identity → verify integrity → state missing data honestly → record recovery event.

16. Platform Adaptation

16.1 Smartphone

- Single-column focus.
- Bottom navigation.
- Full-screen or sheet-based secondary tasks.
- Prominent Capture.
- One review item at a time.
- Progressive disclosure for source, evidence, and history.

16.2 Tablet

- Adaptive sidebar.

- List-detail and source-detail layouts.
- Touch, pointer, keyboard, and drag-and-drop support.
- Compact layouts collapse without losing selection or draft state.

16.3 Desktop

- Persistent sidebar and toolbar.
- Multi-pane workspaces.
- Tables, batch actions, keyboard shortcuts, and resizable panes.
- Detailed import, export, device, and diagnostic administration.

17. Accessibility Requirements

All components must meet UX-0009. At minimum:

- Accurate accessible name, role, value, state, and relationships.
- Logical reading and focus order.
- Visible focus.
- Keyboard and switch operability.
- Platform text scaling and large-text reflow.
- Minimum touch targets.
- Non-color communication.
- Light, dark, and high-contrast validation.
- Reduced-motion behavior.
- Alternatives for hover, drag, camera, QR, voice, and gesture interactions.
- Durable access to information otherwise shown transiently.

18. Content Standards

Component copy must:

- Use plain, direct language.
- Name the object and outcome.
- Avoid blame and alarmism.
- Avoid “magic,” “smart,” or certainty claims that obscure AI limitations.
- Distinguish saved locally from synchronized.
- State data movement honestly.
- Use consistent terms from the Butlerly glossary.
- Support localization without concatenated sentence fragments.

19. State Model Standard

Each stateful component defines:

- Initial state.
- User-triggered transitions.
- System-triggered transitions.
- Asynchronous updates.
- Cancellation.
- Retry.
- Offline behavior.

- Permission loss.
- Provider degradation.
- Authentication failure.
- Success validation.
- Recovery and history.

State changes must not unexpectedly reset focus, discard input, or conceal unfinished work.

20. Analytics Standard

Components may emit privacy-preserving events such as viewed, opened, completed, canceled, failed, retried, deferred, or recovered. Events must not contain record contents, amounts, merchant names, document text, Assistant prompts, encryption material, pairing secrets, or private search terms.

21. Testing Requirements

Each component requires:

- Unit tests for state and data behavior.
- Snapshot or visual regression tests where reliable.
- Accessibility-tree validation.
- Keyboard and focus tests.
- Text scaling and localization tests.
- Light, dark, and high-contrast tests.
- Reduced-motion tests.
- Offline and degraded-state tests where applicable.
- Manual assistive-technology verification.
- Cross-platform parity tests for semantic outcomes.

22. Versioning and Lifecycle

Statuses:

- Proposed.
- Experimental.
- Approved.
- Deprecated.
- Retired.

Semantic versioning applies to the library. Breaking behavior, anatomy, accessibility contract, or data contract requires a major version. Compatible variants and enhancements require a minor version. Corrections require a patch version.

Deprecated components must include replacement guidance and a migration deadline. Product code may not create new uses of deprecated components.

23. Contribution and Approval

A new or changed component proposal must include:

- User problem and evidence.
- Existing components considered.
- Proposed anatomy and behavior.
- Platform adaptations.
- Accessibility contract.
- Privacy and security analysis.
- States and edge cases.
- Engineering feasibility.
- Test plan.
- Migration impact.

Approval requires Design and Engineering. Privacy, Security, AI, or Accessibility review is required when the component affects those domains.

24. Version 1 Component Backlog

P0 Foundations and Controls:

- Buttons, fields, amount/date inputs, selection controls, search, menus, dialogs, sheets, progress, status, and error patterns.

P0 Navigation:

- Bottom navigation, adaptive sidebar, toolbar, tabs, and context path.

P0 Records and Data:

- Record row/card, key-value group, table, source preview, record form, version history, duplicate comparison.

P0 AI:

- Confidence, explanation card, proposal field, consent, action preview, Assistant answer, unavailable state.

P0 Privacy and Security:

- Privacy disclosure, authentication gate, export scope, destructive confirmation, security event.

P0 Synchronization:

- Sync status, device row/detail, pairing, initial sync, conflict comparison, Continue Recent Work.

P1:

- Advanced batch review, rich diagnostics, recovery-package management, complex table customization, and reusable chart patterns.

P2:

- Family/delegated access, cross-platform Android/Windows variants, and advanced selective synchronization.

25. Dependencies

UX-0011 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- UX-0004 — Butlerly Core User Flows.
- UX-0005 — Butlerly Screen and State Inventory.
- UX-0006 — Butlerly AI Interaction and Explainability Specification.
- UX-0007 — Butlerly Design System Foundations.
- UX-0008 — Butlerly Cross-Device Experience and Continuity.
- UX-0009 — Butlerly Accessibility Standard.
- UX-0010 — Butlerly Low-Fidelity Wireframe Specification.

UX-0011 governs:

- High-fidelity product design.
- Native UI implementation.
- Clickable prototypes.
- Engineering component stories.
- Automated component testing.
- Design QA and release acceptance.
- UX-0012 — Usability Testing and UX Metrics.

26. Acceptance Criteria

This specification is complete when:

- Every P0 screen can be assembled from documented components and patterns.
- Each component has a complete behavior, state, platform, accessibility, privacy, and test contract.
- Smartphone, tablet, and desktop renderings preserve equivalent outcomes.
- AI, privacy, synchronization, device trust, error, and recovery components are explicit.
- Engineering can implement components without inventing product behavior.
- Design can maintain consistency without forcing identical layouts across devices.
- Deprecated and changed components have controlled lifecycle rules.
- Traceability exists from journey to flow to screen to wireframe to component to test.

BDS-0000 Conformance

BDS-0000 — Butler Design System v1.0 is the implementation authority for concrete tokens, theme values, typography, spacing/radius, responsive breakpoints, component/state contracts, accessibility implementation, localization/original-data presentation, and Flutter UI constraints. Canonical production component IDs use C-CATEGORY-NNN. Existing BL-* identifiers in this document are legacy descriptive aliases only and must not be used as the authoritative BDS registry key for new implementation or design documentation.

Finance V1 visible primary navigation is Home, Transactions, Review, Search, and Settings; Capture is global/contextual and Assistant is optional/non-primary. Device-trust and

synchronization components are future/optional capability components unless product scope explicitly enables them. Core operation requires no Butlerly account. UI-authored text follows the selected application language; user-entered/imported/scanned/source data preserves its original language/script by default, and original currency/amount remains authoritative with any reference conversion shown separately.